Problem 3 part 2: LBG Algorithm for Quantizing a Two-Dimensional PDF

Problem 3

- Implement the LBG algorithm
  - Approximate the pdf by mass points on a square grid spanning a two-dimensional range with resolution N:

$$x_i = x_0 + \left(\frac{i}{N}\right)(x_N - x_0), y_j = y_0 + \left(\frac{j}{N}\right)(y_N - y_0), \qquad i, j \in \{0, \dots, N\}$$

  - Use several set of initial quantization points to approach global optimum
  - Plot minimum distortion vs trial number
  - Plot optimum quantization points after specified iterations

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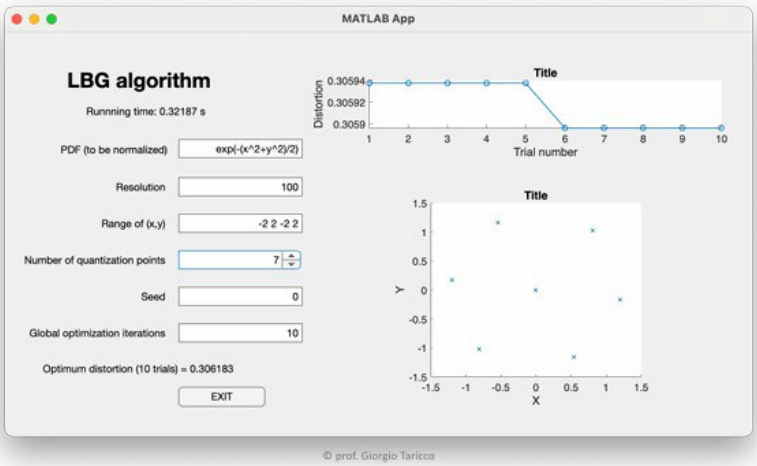
Problem 3



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1 Introduction

In this report, we implement the LBG algorithm to quantize a two-dimensional probability density function (PDF). The PDF is approximated using mass points on a square grid, and we use the LBG algorithm to find the best quantization points that minimize the distortion.

The Linde-Buzo-Gray (LBG) algorithm is a vector quantization method used for data compression and representation. The main goal of the algorithm is to reduce the dimensionality of a dataset by finding a set of representative points that minimize the distortion between the original data points and their quantized versions.

The underlying PDF is given by:

$$P(X, Y) = e^{-\frac{X^2+Y^2}{2}} \tag{1}$$

Then, we normalize the PDF, ensuring that the total probability over the entire 2D space equals 1:

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} P(x, y)\, dx\, dy = 1 \tag{2}$$

2 The LBG Algorithm

We create a grid of points in the two-dimensional space x and y using the following formulas:

$$X_i = \texttt{linspace}(\texttt{minX}, \texttt{maxX}, \texttt{resolution}) \tag{3}$$

$$Y_i = \texttt{linspace}(\texttt{minY}, \texttt{maxY}, \texttt{resolution}) \tag{4}$$

Let the 2D space be represented by $\mathbb{R}^2$, and define the grid points on a square grid:

$$G = \{(x_i, y_i)\}_{i,j} \quad \text{where} \quad i, j \in \{1, 2, \ldots, N\} \tag{5}$$

Here, x_i and y_i represent the coordinates of the grid points in x and y, respectively.

In vector quantization, the goal is to partition the space into regions and assign a codeword to each region. Let the number of codewords be M, where M is the number of quantization points. Initially, we select M random points from the grid:

$$\mathbf{c}_k^{(0)} \quad \text{for} \quad k = 1, 2, \ldots, M \tag{6}$$

These initial centroids $\mathbf{c}_k^{(0)} = (x_k^{(0)}, y_k^{(0)})$ are chosen randomly. This forms a 2D grid of X and Y points using the `meshgrid` function:

$$X, Y \in \mathbb{R}^{\text{resolution} \times \text{resolution}} \tag{7}$$

Each $(X_{i,j}, Y_{i,j})$ corresponds to a grid point. The PDF expression is evaluated over the entire grid, where each point in the grid has a corresponding PDF value $Z_{i,j}$:

$$Z_{i,j} = e^{-\frac{x_i^2+y_i^2}{2}} \tag{8}$$

After evaluation, the PDF is normalized:

$$Z_{i,j} \to \frac{Z_{i,j}}{\sum_{i,j} Z_{i,j}} \tag{9}$$

The normalization ensures that the sum of all PDF values equals 1.

3 Calculating the Distortion

The next step is calculating the distortion to assess how well the quantization approximates the original continuous PDF. The distortion is computed using the formula:

$$D(t) = \sum_{i=1}^{N} \sum_{j=1}^{N} Z(x_i, y_i) \min \left[(x_i - q_{k,x})^2 + (y_i - q_{k,y})^2 \right] \tag{10}$$

This measures the weighted squared distance between each grid point and its nearest quantization point, where the weight is the probability mass at each grid point.

4 Distance Calculation

The distance between a grid point $\mathbf{g} = (g_x, g_y)$ and a centroid $\mathbf{c}_k = (c_{kx}, c_{ky})$ is computed using the Euclidean distance:

$$d(\mathbf{g}, \mathbf{c}_k) = \sqrt{(g_x - c_{kx})^2 + (g_y - c_{ky})^2} \tag{11}$$

For each grid point $\mathbf{g}$, assign it to the centroid $\mathbf{c}_k$ such that:

$$k^* = \arg\min d(\mathbf{g}, \mathbf{c}_k) \tag{12}$$

Once the grid points are assigned to the nearest centroids, the centroids are updated by computing the weighted average of the grid points assigned to each centroid:

$$q_{k,x}^{(t+1)} = \frac{\sum_{(x_i,y_i)\in C_k} x_i Z(x_i, y_i)}{\sum_{(x_i,y_i)\in C_k} Z(x_i, y_i)} \tag{13}$$

$$q_{k,y}^{(t+1)} = \frac{\sum_{(x_i,y_i)\in C_k} y_i Z(x_i, y_i)}{\sum_{(x_i,y_i)\in C_k} Z(x_i, y_i)} \tag{14}$$

The updated position is the weighted centroid of all grid points assigned to each quantization point.

5 Multiple Initializations for Global Optimum

To ensure the algorithm finds the best possible solution and avoids suboptimal results, it is run multiple times with different starting points for the quantization points. At each step, the algorithm tracks the distortion, and the set of quantization points that results in the lowest distortion is chosen as the final solution.

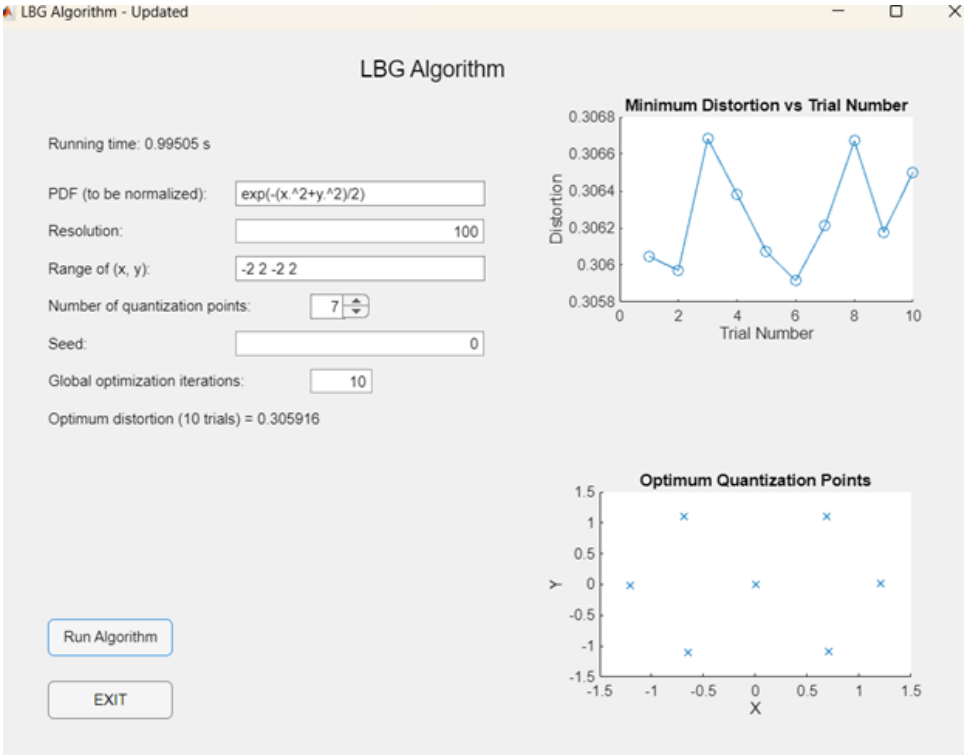
6 Stopping Criterion

The algorithm stops when one of the following conditions is met:

1. A preset number of iterations T is reached.
2. The distortion function stabilizes, i.e., the difference in distortion between two consecutive iterations becomes smaller than a specified threshold ϵ:

$$|D(t+1) - D(t)| < \epsilon \tag{15}$$

7 Results



The PDF is defined as:

$$P(X, Y) = e^{-\frac{X^2+Y^2}{2}} \tag{16}$$

After running the LBG algorithm using the MATLAB GUI, the optimum distortion obtained is 0.305916.

7.1 Minimum Distortion vs Trial Number

The first plot demonstrates the variation in distortion values across 10 trials. The distortion values fluctuate between approximately 0.3058 and 0.3068, indicating consistent but minor variations.

7.2 Optimum Quantization Points

The second plot shows the spatial distribution of quantization points in a two-dimensional plane. These points are symmetrically distributed within the range of $[-1.5, 1.5]$ on both axes, ensuring a balanced quantization scheme.